MindBridge

Agentic AI for Mental Health Awareness & Suicide Prevention

IBM Hackathon — Problem Statement No. 2 | Team: MindBridge

IBM Granite 13B Chat | IBM Watsonx.ai | RAG Pipeline | Python 3.15 | Streamlit

Problem Statement

Mental health has emerged as one of the most pressing yet neglected global health crises of our time. Millions of individuals worldwide silently battle depression, anxiety, post-traumatic stress, and suicidal ideation — conditions that are not only debilitating but life-threatening. Despite the scale of the epidemic, access to timely, affordable, and stigma-free mental health support remains deeply inadequate. Traditional care systems are inherently reactive: people must first recognise their own distress, overcome social stigma, navigate complex healthcare systems, and wait weeks for appointments — by which point a manageable crisis may have escalated into a tragedy. Early warning signals go undetected, coping strategies go unshared, and far too many individuals suffer in silence simply because no one was listening at the right moment.

The absence of intelligent, always-available support creates a dangerous gap between the onset of distress and the delivery of help. Existing digital tools are largely passive — static websites, hotline directories, or generic chatbots that lack the contextual awareness to detect escalating risk or respond with genuine empathy. What is urgently needed is a proactive, agentic AI system that can listen actively, identify distress signals before they reach a breaking point, provide evidence-based coping guidance grounded in a curated mental health knowledge base, and seamlessly connect individuals to human professionals when the situation demands it. By harnessing IBM Granite's large language model, Retrieval-Augmented Generation, and real-time risk prediction — delivered through a compassionate conversational interface — **MindBridge** bridges the critical gap between silent suffering and timely intervention, making mental health support accessible to anyone, anywhere, at any time.

Objectives

1. **Promote Awareness & Education** — Share accessible resources on mindfulness, coping strategies, and self-care through a RAG-powered knowledge base.
2. **Enable Early Detection (RAG)** — Use Retrieval-Augmented Generation to identify distress signals in conversations and journaling inputs in real time.
3. **Provide Empathetic Support** — Build a conversational agent that listens actively and responds with empathy (with clear disclaimers: not a substitute for professional medical advice).
4. **Predict & Prevent Risks** — Forecast stress peaks, suggest preventive strategies, and issue alerts when high-risk signals are detected (with consent).
5. **Connect to Human Support** — Link users to helplines, counselors, or community support for timely intervention.

Proposed Solution

MindBridge is an Agentic AI-powered platform built on **IBM Granite 13B Chat** via **IBM Watsonx.ai** and a Retrieval-Augmented Generation pipeline. A TF-IDF vector store indexes 22 curated mental health articles across 8 categories. Every user message is simultaneously analysed by a **Distress Detection Engine** (6 risk categories, 5 severity tiers: NONE → CRITICAL) and passed to IBM Granite for empathetic, evidence-based response generation grounded in the knowledge base.

A parallel **Risk Predictor** tracks rolling session averages to trigger proactive check-in alerts before crises escalate. On HIGH or CRITICAL detection, verified crisis helplines (India + International) are surfaced instantly alongside coping cards and a Stanley-Brown Safety Plan template. The system is delivered through a **five-tab Streamlit interface** — Chat, Journal, Resources, Insights, and Crisis Help — with interactive mood timeline charts, journal emotional analysis, and optional consent-gated SMS alerts. Every response carries an explicit safe-messaging disclaimer, with ethics embedded at every layer.

Novelty & Uniqueness

Proactive, not reactive — Risk Predictor detects escalating trends across the session and intervenes before a crisis peaks.

RAG-grounded empathy — Every response is anchored in a curated clinical knowledge base, reducing hallucinations.

5-tier real-time risk engine — Continuous scoring across 6 distress categories dynamically shapes tone, content, and urgency.

Journal as detection surface — Analyses free-form writing, not just chat — a pressure-free pathway for those who can't speak their distress.

Ethics by design — Safe messaging, consent-gated alerts, explicit disclaimers, and automatic crisis resource surfacing on every high-risk response.

Future-proof REST architecture — Custom IBM Watsonx REST client works on Python 3.15+, bypassing SDK version constraints.

Technology Stack

Layer	Technology
Large Language Model	IBM Granite 13B Chat `ibm/granite-13b-chat-v2`
Cloud Platform	IBM Watsonx.ai — REST API + IAM Token Authentication
RAG / Vector Search	TF-IDF Vectorizer (scikit-learn) — bigram, cosine similarity
Frontend	Streamlit 1.66 — 5-tab interface (port 8502)
Charts & Visualisation	Plotly — interactive mood timeline & risk distribution
SMS Alerts (optional)	Twilio — consent-gated crisis alerts
Language & Libraries	Python 3.15 · NumPy · scikit-learn · pandas · requests · pydantic

Files Created (13 Files via IBM Bob)

File	Purpose

`app.py`	Main Streamlit UI — 5 tabs
`agent.py`	MindBridgeAgent — session orchestration
`rag_pipeline.py`	TF-IDF vector store + IBM Granite RAG
`watsonx_client.py`	IBM Watsonx REST client (IAM + generation)
`distress_detector.py`	Risk scoring — 6 categories, 5 tiers, rolling predictor
`crisis_manager.py`	Helplines, coping cards, safety plan, SMS
`knowledge_base.py`	22 curated mental health articles (8 categories)
`config.py`	Credentials, thresholds, helplines, keywords
`validate.py`	Component validation suite — 7/7 tests passing
`requirements.txt`	Python dependencies
`.streamlit/config.toml`	Streamlit server config
`.env.example`	Environment variable template
`README.md`	Full documentation & architecture